

ESSAOUIRA, Morocco

WMO: 602200

Lat: 31.5096N

Lon: 9.7726W

Elev: 8

StdP: 101.23

Time Zone: 1.00 (E01)

Period: 90-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.2	9.8	2.0	4.4	16.1	3.8	5.0	14.0	13.5	16.8	11.7	15.8	2.2	70	0.454

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.5	32.9	21.3	29.9	20.5	27.2	19.4	22.4	28.9	21.6	26.1	20.9	24.1	5.8	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.1	15.7	23.0	20.4	15.1	22.3	19.9	14.7	21.8	65.6	29.2	62.7	26.0	60.5	24.3	30.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.8	10.3	9.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	18.3	15.1	15.4	16.5	17.1	18.7	20.3	21.0	21.1	20.9	19.6	17.8	16.0	
	DBStd	3.11	1.98	1.98	2.42	2.00	2.03	2.15	2.86	2.62	2.17	2.09	2.24	1.99	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	
	HDD18.3	440	101	84	69	47	16	1	2	0	1	8	35	77	
	CDD10.0	3036	159	151	201	214	270	310	342	344	326	298	235	186	
	CDD18.3	435	2	2	12	11	27	61	86	86	77	48	20	4	
	CDH23.3	2014	11	7	61	50	118	262	539	488	289	122	64	3	
	CDH26.7	794	1	0	13	17	41	103	262	229	91	30	8	0	
Wind	WSAvg	4.6	3.7	4.7	4.6	5.1	5.0	5.3	5.5	5.3	4.6	3.9	3.9	3.9	
Precipitation	PrecAvg	303	47	39	34	30	9	2	0	1	4	28	49	52	
	PrecMax	456	175	153	124	107	35	16	4	26	27	126	203	168	
	PrecMin	124	0	1	0	1	0	0	0	0	0	0	0	0	
	PrecStd	96	46	35	29	28	9	4	1	5	7	36	52	50	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.8	24.1	28.1	27.8	32.1	34.3	36.9	37.0	32.2	30.9	27.9	23.8	
		MCWB	14.7	14.3	16.8	18.1	19.8	24.0	22.3	22.6	20.8	18.1	17.4	15.6	
	2%	DB	22.0	21.9	24.8	23.8	26.2	28.9	32.5	31.9	28.9	26.2	25.1	21.4	
		MCWB	15.3	15.9	15.8	16.6	18.4	20.3	21.8	21.5	20.5	18.6	17.5	17.1	
	5%	DB	20.3	20.2	22.2	21.1	23.1	25.8	29.0	28.1	26.5	24.0	22.9	20.3	
		MCWB	16.1	16.1	16.1	17.1	17.7	19.5	20.9	20.8	19.4	19.2	18.1	17.4	
	10%	DB	19.1	18.8	20.2	19.9	21.1	22.9	25.1	24.3	23.8	22.4	21.2	19.4	
		MCWB	15.9	15.7	16.4	17.3	18.7	19.3	19.6	20.1	19.6	19.3	17.7	17.0	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.4	19.2	20.0	20.0	21.1	23.3	23.3	23.1	23.1	21.8	21.2	19.6
			MCDB	20.7	21.0	22.8	21.4	24.0	29.6	32.6	32.8	27.6	24.0	23.5	20.7
2%		WB	18.4	18.0	18.8	19.0	20.1	21.4	22.3	22.1	21.7	20.9	20.1	18.8	
		MCDB	20.0	19.6	21.1	20.6	22.4	24.5	30.1	28.6	25.1	23.0	22.4	20.1	
5%		WB	17.2	17.0	17.9	18.3	19.3	20.5	21.2	21.2	20.8	20.1	19.2	18.2	
		MCDB	19.1	18.7	20.0	20.0	21.6	23.2	26.0	25.8	23.1	22.3	21.4	19.5	
10%		WB	16.2	16.1	17.0	17.6	18.7	19.8	20.4	20.5	20.2	19.6	18.3	17.4	
		MCDB	18.2	18.1	19.2	19.4	20.9	22.2	23.8	23.6	22.5	21.7	20.4	18.9	
Mean Daily Temperature Range	5% DB	MDBR	7.9	7.0	7.6	5.9	5.5	5.8	6.0	5.5	5.5	5.0	6.5	6.1	
		MCDBR	10.7	9.7	11.2	9.5	10.0	11.4	14.1	13.0	11.3	8.8	10.0	7.2	
		MCWBR	7.1	6.1	6.8	5.3	5.0	4.5	5.2	4.5	4.7	4.5	6.1	5.5	
	5% WB	MCDBR	6.6	6.3	7.5	6.4	7.2	7.0	10.5	10.0	7.8	6.0	6.5	5.6	
		MCWBR	5.8	5.1	6.0	4.6	4.6	4.2	4.5	4.3	4.3	4.0	5.1	4.8	
Clear-Sky Solar Irradiance	taub	0.329	0.364	0.416	0.413	0.434	0.469	0.531	0.526	0.448	0.412	0.370	0.345		
	taud	2.509	2.398	2.213	2.259	2.217	2.123	1.940	1.977	2.212	2.313	2.417	2.475		
	Ebn at Noon	898	892	863	880	859	825	773	771	822	827	839	856		
	Edh at Noon	87	107	138	137	144	158	188	179	136	115	95	87		
All-Sky Solar Radiation	RadAvg	3.52	4.40	5.66	6.72	7.43	7.77	7.77	7.05	6.00	4.56	3.62	3.15		
	RadStd	0.19	0.29	0.26	0.29	0.34	0.25	0.22	0.20	0.20	0.22	0.23	0.22		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (4 neighbors)	+0.24	N/A	N/A	N/A	-0.32	N/A	N/A	N/A	+153	+171

Nomenclature: See separate page